Supplementary Material: Calibration Transfer and Updating in Educational Risk Prediction: A Cross-Cohort Reliability Audit

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ABSTRACT This supplement details features, sample selection, absolute metrics, implementation and sensitivity analyses. Original-label, user-assignment, future-inactivity and later-presentation updating analyses have distinct outcomes or supported populations. Target OOF calibration is retrospective; recent-label updating is tested on a later presentation. Original supplied-label timing remains unverified.

INDEX TERMS Calibration transfer, calibration updating, educational data mining, probability calibration.

S1. LANDMARK-SPECIFIC FEATURE INVENTORY

OULAD appends source-defined course, progress-state, and course-by-progress indicators to its 46 numeric predictors. Progress states are no_due, due_complete, and due_missing. XuetangX retains actions with nonzero source-training counts; other actions contribute to OTHER. No post-landmark event or assessment score enters these features.[1]

S2. SECONDARY LOG-LOSS ANALYSIS

All six $A^{\log}$ values were positive, supporting the qualitative Brier ordering. Log loss is sensitive to extreme probabilities; the fixed 10^{-6} clipping rule applies.[2]

S3. FULL-PIPELINE USER-ASSIGNMENT SENSITIVITY

Within the original eligible course pool, every username was assigned to its earliest observed base course before source role allocation or fitting, with lexical tie-breaking. This retained 62,142 of 116,363 eligible enrollments and restricted each retained user to at most one eligible base course. Of 60,446 eligible users, 32,363 originally occurred in multiple base courses. Predictors, vocabulary, preprocessing, source calibrators, and target OOF calibrators were all refitted with the frozen settings. The allocation is outcome-blind conditional on an outcome-dependent eligible pool.[3]

The analysis completed 908 jobs with no recorded fitting failures. All six refit-minus-original intervals included zero; this does not establish equivalence. Contrasts combine changes in membership and fitting because identical jobs do not imply identical target learners. Original full-sample intervals retain their crossed-user-dependence limitation. This full refit supersedes the weaker evaluation-only filter as the principal repeated-user sensitivity; the earlier filter did not refit any model.

S4. ANALYTIC SAMPLE AND ABSOLUTE TARGET PERFORMANCE

The audit identified 84 adjacent same-course candidate pairs; 67 passed pair-level requirements. Every one of the 17 exclusions involved insufficient original label-class support: 14 failed only the 50-per-class rule, two also failed 300 enrollments, and one also failed the seven-day gap. Landmark rules require 50 unseen active target learners and a target minority class of 10, alongside source-role class support. The primary analysis has 48/57/66 pairs and 218/261/312 pair-seed combinations at Days 7/14/28. After full user assignment, original-label support is 19/36/44 pairs (85/158/211 pair-seeds); the new-outcome analysis supports 35/50/55 pairs (151/216/243 pair-seeds). These are three separate selection branches. Not every pair retains all five seeds.[4] The source training minority class must contain at least 5 learners, validation at least 2, and source calibration at least 10 within at least 30 total calibration learners. Both supplemental refits retain the original 67-pair pool and reapply these landmark-level source and target rules; the 300-enrollment and 50-per-class pair-level filters are not reapplied after user assignment or relabeling.

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TABLE S1. Feature families and time boundaries. Exact names, transformations, and code locations are supplied in docs/feature_schema.csv in the reproducibility materials.

Platform / family	Included information	Time and fitting boundary
OULAD: VLE volume and timing	Click volume, active days, sites/types, recent versus prior-week activity, recency, zero-activity indicators	Day 0 through landmark; registered, not yet withdrawn risk set
OULAD: VLE type counts	Counts by VLE activity type	Only events observed by landmark
OULAD: assessment and schedule	Known non-exam deadlines and weights; observed submissions; completion, lateness and overdue counts	Submission timestamps no later than landmark; no assessment score
OULAD: contextual indicators	Course, progress state, and their interactions	Source-defined categorical encoding
XuetangX: engagement summaries	Volume, active days, sessions, objects, timing, recency, within-window change and weekend fraction	First observed course event through landmark; no future event
XuetangX: action counts/fractions	Counts and fractions for source-training actions; OTHER for remaining actions	Vocabulary fitted only on source training learners

TABLE S2. Pair-macro log-loss gains relative to raw probabilities. No confirmatory intervals were computed for this secondary metric.

Day	Model	Pairs	$G_T^{\log}$	$G_T^{*\log}$	$A^{\log}$
7	LR	48	0.1887	0.3483	0.1596
7	LightGBM	48	-0.0015	0.0784	0.0799
14	LR	57	0.1111	0.2184	0.1073
14	LightGBM	57	-0.0014	0.0757	0.0771
28	LR	66	0.0493	0.1110	0.0617
28	LightGBM	66	0.0016	0.0500	0.0484

TABLE S3. Original-label full refit on the common job panel. Difference intervals resample base courses jointly; target learner membership differs between original and refit.

Day	Model	Pairs / clusters	Jobs / original	Original A	Refit A	ΔA	95% CI: ΔA
7	LR	19 / 19	85 / 218	0.0370	0.0458	0.0088	[-0.0098, 0.0306]
7	LightGBM	19 / 19	85 / 218	0.0264	0.0323	0.0059	[-0.0090, 0.0247]
14	LR	36 / 29	158 / 261	0.0353	0.0411	0.0058	[-0.0023, 0.0152]
14	LightGBM	36 / 29	158 / 261	0.0278	0.0276	-0.0002	[-0.0066, 0.0068]
28	LR	44 / 35	211 / 312	0.0178	0.0172	-0.0006	[-0.0072, 0.0042]
28	LightGBM	44 / 35	211 / 312	0.0169	0.0148	-0.0020	[-0.0072, 0.0024]

TABLE S4. Primary XuetangX absolute target metrics, averaging seeds within pair and then pairs equally. Target OOF is retrospective.

Day	Model	Pairs	Truth=1	Raw AUROC	Raw BS	Transferred BS	Target OOF BS
7	LR	48	0.7358	0.6184	0.2080	0.1844	0.1432
7	LightGBM	48	0.7358	0.6349	0.1716	0.1718	0.1411
14	LR	57	0.7514	0.6630	0.1814	0.1685	0.1342
14	LightGBM	57	0.7514	0.6841	0.1574	0.1581	0.1303
28	LR	66	0.7623	0.7259	0.1488	0.1451	0.1264
28	LightGBM	66	0.7623	0.7459	0.1381	0.1379	0.1213

Original truth timing and source-label availability remain unverified. Transferred and target OOF Brier scores refer to the same primary target observations. The new endpoint described in S7 is a separate behavioral outcome, not a repair of the original truth label.

S5. IMPLEMENTATION AND SCORING

The five seeds are 17, 29, 43, 59, and 71. Full source calibration fits the historical transfer map; source and target OOF gains use five learner-disjoint folds. OULAD pools source calibration learners across eligible courses within season, day, model, and seed; XuetangX calibrates within each source presentation. Original-label source roles are fixed across landmarks within presentation/seed, including after user assignment. Future-outcome labels vary by landmark, so their stratified source roles are regenerated separately for each landmark. The Platt penalty is 10^{-5} around identity parameters, with slope bounded to [0,30] and intercept to [-

TABLE S5. Base-predictor and calibrator settings. Source-validation criteria differ between the development and distinct-platform protocols.

Setting	OULAD	XuetangX
LR	C ∈ {0.1, 1.0}; source-validation Brier selection; L-BFGS; 1,500 iterations	C = 1; fixed; L-BFGS; 800 iterations
LightGBM	7 or 15 leaves; 350 trees; rate 0.03; min child 40; L2 = 5; source-validation Brier selection	15 leaves; max depth 5; 180 trees; rate 0.05; min child 20; L2 = 1; validation-log-loss early stopping, patience 20
Logistic preprocessing	Signed log1p; scaler fit on source training only	Signed log1p; scaler fit on source training only
Calibration folds	Five StratifiedGroupKFold folds by student ID for source and target OOF	Five stratified learner-disjoint folds within source and target presentations
Platt / clipping	Probabilities clipped to [10^{-6}, 1-10^{-6}]; penalized Bernoulli log loss, identity-centered penalty 10^{-5}	Same clipping and Platt objective; source full fit transferred unchanged

30,30]. Brier gains are rounded from full precision; rounded displayed gains may differ from their displayed subtraction by 0.0001.[5]

For label-budget matching, each target calibration-training fold used $\min(n_{S,\mathrm{cal}}, n_{T,\mathrm{train}})$ observations, sampled without replacement in three fixed repeats. Samples lacking either class were redrawn up to 100 times; held-out test folds were unchanged. Sampling therefore conditions on class support. Beta calibration used $\mathrm{sigmoid}(a\log p - b\log(1-p) + c)$, with $a, b \in [0, 30]$, $c \in [-30, 30]$, and a 10^{-5} identity-centred quadratic penalty. SciPy L-BFGS-B used initial parameters $(1, 1, 0)$ and at most 300 iterations. Isotonic regression was increasing, with bounds 10^{-6} and $1 - 10^{-6}$, and clipped out-of-range inputs.

The DFMC comparator used the documented local LightGBM implementation of the original squared-loss tree-correction algorithm, with depth 2, four leaves, learning rate 0.01, bagging fraction 0.5 each round, minimum leaf size 20, at most 2,000 rounds, and 50-round early stopping on a stratified 30% inner validation set. Inputs comprised raw probability and indicators for all learners, activity at or above the source-calibration median, any video, any problem, and any forum activity. Raw outputs were clipped for scoring, and out-of-range rates were retained. All five predictor seeds were included. Thus, "fixed-default DFMC" refers to these frozen implementation settings, rather than unspecified library defaults.

Executed dependencies were Python 3.12.10, NumPy 2.5.3, pandas 3.0.6, SciPy 1.18.1, scikit-learn 1.9.1, LightGBM 4.7.0, and PyArrow 25.0.1. Fixed random seeds, remaining tree settings, the retained upstream license, original implementation hashes, and distributed-file hashes are documented in the reproducibility materials. Logistic full Platt used L-BFGS-B with initial $(1, 0)$, maximum 300 iterations and tolerance 10^{-10}; calibration coefficient estimates are bounded, regularized diagnostics.

S6. EXPANDED ROBUSTNESS ESTIMATES

The exploratory source-observable gate for A > 0.001 yielded leave-one-base-course-out AUROC 0.594 for a logistic meta-model and 0.610 for a shallow tree, against approximately 87% positive prevalence. It is a failed screening diagnostic. The complete random-source contrast is reported in Table S10; LR Days 14 and 28 have difference intervals spanning zero. DFMC is a fixed-default robustness comparator rather than the proposed method. These exploratory analyses were not multiplicity-adjusted.[6]

S7. LABEL PROVENANCE AND FUTURE SEVEN-DAY LOGGED INACTIVITY

The supplied archive contains train/train.csv, test/test.csv, and model1.model but no original label generator, codebook, or label timestamp. The observed source-last-event to target-first-event gap across the 67 pairs has minimum 24.04 days and median 123.02 days. These gaps do not establish original truth availability. All 67 source and target observed spans extend through Day 35, but observed last events do not prove complete provider data collection.[7]

$$Y_{i,d}^{(7)} = \mathbf{1}\{N_i(t_c + d, t_c + d + 7] = 0\}. \tag{1}$$

Here t_c is the first observed event time of the presentation, and N_i counts learner i's logged events in the specified interval. Features use events through the landmark inclusively; the new outcome uses events strictly after the landmark and through Day d+7 inclusively. Under nominal observed coverage, source outcome windows end at least 46.01/39.01/25.01 days before the target first event for Days 7/14/28. This endpoint is short-term logged inactivity, not certified dropout. Target adaptation still requires target outcomes, and the candidate pool inherits original truth-based eligibility.

Before user assignment, agreement between supplied truth and the new outcome among active enrollments in eligible courses was 0.7268 (45,532 enrollments), 0.7631 (70,893), and 0.8255 (102,384). These agreement rates do not recover the original label definition. Neither endpoint sensitivity is an independent platform replication or a prospective policy evaluation.

S8. OULAD IDENTITY AUDIT AND STRATIFIED ESTIMATES

Exclusion covers all modules in the full source-season presentation: 8,830 IDs in 2013J and 4,679 in 2013B. The B

[5]Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

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[7]Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

TABLE S6. Adaptation gaps in robustness families. Populations and pair counts differ; these are not treatment-effect contrasts.

Analysis	Day	Pairs	A: LR	A: LightGBM
Primary Platt	7	48	0.0412	0.0307
Primary Platt	14	57	0.0343	0.0277
Primary Platt	28	66	0.0188	0.0166
Matched target labels	7	48	0.0392	0.0287
Matched target labels	14	57	0.0326	0.0260
Matched target labels	28	66	0.0173	0.0151
Beta	7	48	0.0425	0.0310
Beta	14	57	0.0349	0.0276
Beta	28	66	0.0200	0.0174
Isotonic	7	48	0.0494	0.0352
Isotonic	14	57	0.0415	0.0318
Isotonic	28	66	0.0226	0.0196
Fixed-default DFMC	7	48	0.0494	0.0274
Fixed-default DFMC	14	57	0.0451	0.0276
Fixed-default DFMC	28	66	0.0229	0.0172
All-enrollment refit	7	67	0.0251	0.0213
All-enrollment refit	14	67	0.0232	0.0209
All-enrollment refit	28	67	0.0183	0.0154
Earliest-to-latest	7	18	0.0308	0.0171
Earliest-to-latest	14	23	0.0331	0.0163
Earliest-to-latest	28	23	0.0143	0.0088

TABLE S7. Future seven-day logged-inactivity results after full-pipeline user assignment (1,220 jobs). Each row uses its own supported course pairs.

Day	Model	Pairs / clusters	G_S	G_T	G_T^*	A	95% CI for A
7	LR	35 / 28	0.0152	0.0258	0.0456	0.0197	[0.0135, 0.0268]
7	LightGBM	35 / 28	0.0025	-0.0047	0.0098	0.0146	[0.0094, 0.0203]
14	LR	50 / 38	0.0084	0.0217	0.0548	0.0331	[0.0211, 0.0486]
14	LightGBM	50 / 38	-0.0010	-0.0021	0.0158	0.0179	[0.0118, 0.0256]
28	LR	55 / 38	0.0046	0.0097	0.0331	0.0235	[0.0147, 0.0338]
28	LightGBM	55 / 38	-0.0002	-0.0004	0.0137	0.0140	[0.0094, 0.0193]

target additionally excludes retained J-development learners. Using studentInfo.csv, frozen roles and saved predictions, all 80 model jobs had zero target overlap with source training, validation, pooled calibration, their union, or the full source presentation. Source roles were mutually disjoint, and all 800 OOF fit–evaluation checks passed. A separate reproducibility run rebuilt all 80 jobs from prepared features: 360 primary and 1,080 matched-budget result rows agreed within 10^{-15}; mechanism classifications agreed exactly. The reproducibility materials contains aggregate audit counts and hashes without learner IDs.[8]

Table S8 averages the five seeds within course–landmark–model cells and then weights course pairs equally within each season. These are descriptive strata with only six J pairs and three B pairs, not independent replicate estimates.

[8]Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

S9. HISTORICAL TRANSFER UNCERTAINTY AND PAIR DISTRIBUTIONS

The same 2,000-draw base-course-cluster bootstrap used for the primary adaptation gap also estimates uncertainty in transfer gain. Seeds are first averaged within pair, and repeated pairs within the same base course are resampled together. Table S9 additionally reports the empirical distribution of pair-level transfer gains. These intervals retain the primary analysis's crossed-user-dependence limitation. Complete intervals for all three gains and all pair-level aggregate values are available in the reproducibility materials.[9]

S10. RANDOM HISTORICAL-SOURCE CONTROL

For each target job, the historical calibrator is replaced by a map fitted on an earlier unrelated base course while the target raw predictions and outcomes stay fixed. Eligible sources have a last observed event before the target first event and a different provider–base-course key. Twenty histories

[9]Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

TABLE S8. OULAD season–landmark–model Brier gains. J and B denote the 2013-to-2014 presentation seasons.

Season	Day	Model	Pairs	G_S	G_T	G_T^*	A
J	7	LR	6	0.00001	-0.00014	-0.00010	0.00003
J	7	LightGBM	6	-0.00006	-0.00034	-0.00012	0.00021
J	14	LR	6	-0.00031	0.00005	0.00132	0.00127
J	14	LightGBM	6	-0.00025	0.00008	0.00104	0.00096
J	28	LR	6	-0.00019	0.00032	0.00219	0.00188
J	28	LightGBM	6	-0.00026	-0.00002	0.00065	0.00067
J	56	LR	6	-0.00013	-0.00008	0.00033	0.00041
J	56	LightGBM	6	-0.00015	0.00012	0.00160	0.00147
B	7	LR	3	-0.00036	-0.00000	0.00310	0.00310
B	7	LightGBM	3	-0.00055	0.00034	0.00509	0.00476
B	14	LR	3	0.00004	0.00069	0.00542	0.00473
B	14	LightGBM	3	0.00021	0.00016	0.00542	0.00526
B	28	LR	3	0.00002	-0.00014	0.00243	0.00257
B	28	LightGBM	3	0.00041	0.00050	0.00420	0.00370
B	56	LR	3	-0.00010	-0.00019	0.00407	0.00426
B	56	LightGBM	3	0.00017	-0.00024	0.00376	0.00401

TABLE S9. Primary XuetangX transfer gains and across-pair heterogeneity. Positive counts use seed-averaged pair gains.

Day	Model	Positive / pairs	Mean G_T	95% CI	Median	[Q1, Q3]
7	LR	45 / 48	0.0237	[0.0177, 0.0306]	0.0205	[0.0090, 0.0368]
7	LightGBM	17 / 48	-0.0002	[-0.0040, 0.0050]	-0.0023	[-0.0053, 0.0014]
14	LR	51 / 57	0.0130	[0.0066, 0.0187]	0.0123	[0.0036, 0.0202]
14	LightGBM	29 / 57	-0.0007	[-0.0034, 0.0015]	0.0000	[-0.0026, 0.0018]
28	LR	55 / 66	0.0036	[-0.0009, 0.0073]	0.0028	[0.0009, 0.0088]
28	LightGBM	30 / 66	0.0002	[-0.0010, 0.0016]	-0.0004	[-0.0023, 0.0009]

are drawn with replacement and averaged within job, then seeds within pair. Intervals condition on these realized source draws. The bootstrap resamples base courses jointly for both mappings and their paired difference. Eligibility, random seeds, and source-choice code are provided in the reproducibility package. Historical maps arise from separately fitted source predictors; the comparison tests this implemented control rather than a source-course effect isolated from score-scale differences.[10] Both constituent transfer-gain intervals are included in the machine-readable results. Same-course mean gain is higher in all six rows; the LR Day 14 and Day 28 difference intervals span zero.

S11. SLOPE-ONLY AND SUPPORTING CALIBRATION DIAGNOSTICS

All three logistic calibration structures use the same source and target fold partitions. Slope-only results are supporting analyses: the frozen primary-method list specified intercept-only and full Platt. The slope-only map fixes the intercept at zero; the intercept-only map fixes slope at one; full Platt estimates both. Table S11 reports their target OOF gains, while the machine-readable structure table also contains source-transfer gains and their intervals. These comparisons describe predictive utility rather than mutually exclusive or causal components.[11] The fixed-slope calibration intercept (calibration-in-the-large) is fitted with the raw logit as an offset; the joint calibration slope comes from the bounded, mildly regularized two-parameter full-pool fit defined in S5. Target quantities use observed target outcomes and are descriptive diagnostics. Table S12 reports across-pair medians and interquartile ranges after averaging seeds. ECE uses ten approximately equal-count bins after stable sorting within each job, with sample-size-weighted absolute observed-minus-predicted differences; tied scores may be split across bins. It is a supporting, bin-dependent description and did not select models or establish the primary conclusions.

S12. FOLDWISE TARGET-PREVALENCE BASELINE

For held-out target fold F_k, define[12]

$$\hat{p}_{i,T}^{\text{null}} = \frac{1}{|T \setminus F_k|} \sum_{j \in T \setminus F_k} y_j, \qquad i \in F_k. \tag{2}$$

The saved target Platt folds are reused exactly. No held-out outcome enters its own constant prediction. Brier losses are computed within each job before averaging seeds and course pairs, avoiding substitution of a globally averaged prevalence.

[10] Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

[11] Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

[12] Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

TABLE S10. Same-course versus random earlier unrelated-course historical calibration. Paired difference is same minus random transfer gain.

Day	Model	Pairs / jobs	Same G_T	Random G_T	Difference	95% CI: difference
7	LR	48 / 218	0.0237	0.0121	0.0115	[0.0048, 0.0189]
7	LightGBM	48 / 218	-0.0002	-0.0137	0.0135	[0.0090, 0.0191]
14	LR	57 / 261	0.0130	0.0101	0.0029	[-0.0024, 0.0084]
14	LightGBM	57 / 261	-0.0007	-0.0067	0.0061	[0.0028, 0.0090]
28	LR	66 / 312	0.0036	0.0010	0.0027	[-0.0012, 0.0062]
28	LightGBM	66 / 312	0.0002	-0.0041	0.0044	[0.0031, 0.0055]

TABLE S11. Target OOF Brier gains under intercept-only, slope-only and full-Platt restrictions.

Day	Model	Pairs	Intercept	Slope	95% CI: slope	Full Platt
7	LR	48	0.0358	0.0267	[0.0113, 0.0446]	0.0649
7	LightGBM	48	0.0281	0.0161	[0.0045, 0.0334]	0.0305
14	LR	57	0.0302	0.0167	[0.0070, 0.0286]	0.0472
14	LightGBM	57	0.0250	0.0113	[0.0043, 0.0211]	0.0271
28	LR	66	0.0135	0.0094	[0.0049, 0.0148]	0.0224
28	LightGBM	66	0.0143	0.0066	[0.0035, 0.0104]	0.0169

TABLE S12. Calibration diagnostics: median [Q1, Q3] over course pairs. Intercepts and slopes diagnose raw probabilities; target raw ECE is reported in the last column.

Day	Model	Source intercept	Target intercept	Source slope	Target slope	Target ECE
7	LR	-0.040 [-0.124, 0.071]	0.755 [0.165, 1.471]	0.445 [0.312, 0.574]	0.202 [0.120, 0.261]	0.167 [0.126, 0.229]
7	LightGBM	-0.072 [-0.153, 0.026]	0.459 [0.056, 0.843]	1.086 [0.873, 1.299]	0.692 [0.483, 1.281]	0.110 [0.084, 0.161]
14	LR	-0.010 [-0.122, 0.121]	0.452 [-0.157, 1.300]	0.534 [0.370, 0.667]	0.297 [0.191, 0.414]	0.142 [0.101, 0.168]
14	LightGBM	-0.052 [-0.140, 0.026]	0.226 [-0.398, 0.696]	1.097 [0.957, 1.214]	0.924 [0.630, 1.110]	0.090 [0.071, 0.129]
28	LR	-0.005 [-0.086, 0.064]	0.130 [-0.369, 0.669]	0.628 [0.507, 0.765]	0.445 [0.343, 0.577]	0.092 [0.063, 0.140]
28	LightGBM	-0.052 [-0.127, -0.002]	-0.003 [-0.445, 0.408]	1.071 [0.944, 1.187]	0.930 [0.769, 1.128]	0.068 [0.052, 0.110]

TABLE S13. Target Platt versus a foldwise target-prevalence predictor. The paired benefit is null Brier loss minus Platt Brier loss.

Day	Model	Pairs	Raw BS	Null BS	Platt BS	Benefit	95% CI: benefit
7	LR	48	0.2080	0.1481	0.1432	0.0049	[0.0026, 0.0075]
7	LightGBM	48	0.1716	0.1481	0.1411	0.0070	[0.0044, 0.0098]
14	LR	57	0.1814	0.1450	0.1342	0.0108	[0.0073, 0.0148]
14	LightGBM	57	0.1574	0.1450	0.1303	0.0146	[0.0104, 0.0190]
28	LR	66	0.1488	0.1481	0.1264	0.0217	[0.0161, 0.0281]
28	LightGBM	66	0.1381	0.1481	0.1213	0.0268	[0.0202, 0.0339]

All 1,582 primary jobs and 7,910 target folds were checked against the original fold generator; recalculated stored Platt Brier metrics agreed to below 10^{-15}. Base-course-cluster resampling uses the same paired procedure as the primary gains. Both this comparator and target Platt require target outcomes retrospectively. All six mean Platt benefits over the constant predictor have positive intervals. At Day 7, the constant predictor already improves on raw scores, while Platt supplies a smaller additional gain. By Day 28, LightGBM's raw scores also outperform the constant baseline. Individual scores retain information beyond the event rate; this is not a causal decomposition.

S13. REPRODUCIBILITY MATERIALS

The analysis code is available at https://github.com/13406330468/Educational-Risk-Prediction, with analysis-source commit `538ebd04da2716a9725354be5718542e2f0ce179`. The separate `Frozen_Results.zip` archive contains the aggregate-results container and numeric evidence registry. It must

be extracted into the repository root before executing the documented aggregate-verification commands. Its SHA-256 digest is `22ad313f237dd62b8e86a09eaf87cbc3f6aaa18467e6cd76a525252dc1ebd4e3`. These source and archive identifiers define the analysis materials independently of subsequent repository changes.[13]

Fresh preparation from the original XuetangX ZIP and 7,610 model fits reproduce the key two- and three-presentation time-defined pipelines. This is an independent execution of the same algorithms, not an independent implementation or team. It does not imply a complete rerun of OULAD or every historical analysis. Frozen-aggregate regeneration and raw-data recomputation remain separate commands. No learner records or individual predictions are distributed. Section S24 records the executed checks and numerical tolerances.

Original supplied-label timing remains unresolved. The time-defined endpoint does not authenticate it. Author information, funding, ethics and AI assistance are disclosed in the main manuscript.

S14. PRIMARY SAMPLE SUPPORT

Table S14 describes the source calibration and target populations used by both models. Source quantities average the available fixed seeds within each pair. Target learners do not change with model seed, but different supported pairs remain at each landmark. Full ranges and exact per-job class counts are in the aggregate evidence. These summaries should not be interpreted as a common population across days.[14]

S15. FUTURE OUTCOME WITHOUT ORIGINAL-LABEL PAIR SELECTION

We restart from all 84 adjacent course pairs and retain 81 with at least 300 enrollments per presentation and a seven-day observed gap. The selector reads no original-label columns. Enrollment identifiers are joined to seven-day future event indicators computed from all 42,110,402 archived events. Absence of any event in $(d, d+7]$ defines the new outcome; observations up to and including d supply features. This reconstructs a distinct endpoint without claiming dropout or verified complete logging.[15]

For each day and source seed, 60/20/20 roles are regenerated using the new outcome. We require source train minority count at least five, validation minority at least two, calibration N at least 30 and minority at least ten; the active, source-unseen target requires N at least 50 and minority at least ten. Both observed presentation spans must cover $d+7$. The original-label class thresholds are never reused. Supported pair counts are 73/79/80, with 339/368/386 source splits per model (2,186 jobs). At least one day supports 13 of the 14 structurally eligible pairs absent from the original cohort. Failures in an admissible job abort execution; unsupported units remain in the selection table. This changes pairing selection and endpoint, but does not impose cross-course user assignment, allowing the original identity restriction and the new pool to remain separate sensitivity questions.

S16. THREE-PRESENTATION TEST AND RECENT-LABEL BUDGETS

The 81 structural pairs form 33 consecutive triples. Source fitting uses the same roles, hyperparameters, train-only vocabulary and preprocessing as the original pipeline, with the new future-inactivity label. A single fitted source predictor scores both T1 and T2. Source users are excluded from T1; users present anywhere in either earlier presentation are excluded from T2, including users absent from active fitting subsets. Both earlier endpoint windows must end strictly before the next prediction landmark. The minimum supported lead is 56.07 days for both S-to-T1 and T1-to-T2 checks. This verifies timestamp ordering in the archive, not real-time data arrival.[16]

Each source split requires the S/T1 pair support in S15, and T2 requires at least 50 active unseen learners and ten observations in each class. There are 141, 152 and 155 supported source splits at Days 7, 14 and 28, each fitted with both base models. The 30/32/32 supported triples belong to 23/24/24 base courses. Nested T1 samples follow a random permutation without outcome stratification; fitting requires at least ten labels and both classes. Unsupported draws are retained and are not replaced. Day 7 has ten unsupported model–seed budget evaluations at 10%; all other budgets are supported. Median available T1 counts are 217, 354 and 583 across supported source splits, with ranges 93–2,970, 161–3,317 and 257–3,577. The realized fraction uses floor(fraction times T1 N).

At full budget, the updated map improves on the recent constant by 0.0207–0.0392 Brier units across the six conditions; all six pointwise intervals are positive. Table S16 gives every budget and its actual supported population. The constant recent-label comparator uses the exact T1 subset prevalence and never T2 outcomes. Inference averages seeds within triple, then triples equally. Future-update tables use the raw-data baseline execution and 100,000 course-cluster resamples. For Figure 5, source splits supported at every fraction are fixed before plotting all budgets. These displayed intervals are pointwise. Section S21 supplies the multiplicity sensitivity; neither analysis establishes a confirmatory update policy. This new endpoint and triple population differ from the original-label primary pairs, so their magnitudes cannot estimate a common treatment effect.

S17. EQUAL-BUDGET WITHIN-PRESENTATION CONTROL

For each saved primary predictor, set $m = \min(n_{S,cal}, \lfloor n_T/3 \rfloor)$. Randomly draw m source calibration observations. Independently permute target users into two disjoint calibration groups H and C, each of size m, and an

[13] Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.
[14] Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.
[15] Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.
[16] Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

TABLE S14. Primary sample sizes and positive-label counts: median [Q1, Q3] across pairs after averaging seeds. Both base models use the same learner roles, so counts are shared. Decimal counts arise from seed averaging.

Day	Source cal. N	Target N	Source cal. positives	Target positives	Source prevalence	Target prevalence
7	101.5 [61.2, 120.0]	217.0 [144.8, 353.5]	68.8 [39.8, 89.3]	155.5 [103.8, 292.2]	0.726 [0.657, 0.778]	0.822 [0.657, 0.895]
14	134.4 [85.8, 183.8]	359.0 [228.0, 625.0]	106.0 [61.2, 132.2]	269.0 [190.0, 496.0]	0.778 [0.703, 0.818]	0.830 [0.688, 0.892]
28	179.6 [108.6, 252.1]	567.5 [370.0, 889.8]	130.1 [79.3, 188.3]	419.0 [314.2, 667.2]	0.813 [0.731, 0.863]	0.827 [0.703, 0.891]

TABLE S15. Baseline structural-pool inactivity gains, with 100,000-draw pointwise course-cluster intervals for A. P/C denotes supported pairs/clusters. These intervals are not the 12-comparison expansion for the two restricted populations.

Day	Model	P/C	G_S	G_T	G_T^*	A	Pointwise interval
7	LR	73/47	0.0108	0.0185	0.0447	0.0261	[0.0186, 0.0352]
7	LightGBM	73/47	-0.0005	0.0016	0.0190	0.0174	[0.0114, 0.0246]
14	LR	79/48	0.0053	0.0112	0.0369	0.0257	[0.0179, 0.0347]
14	LightGBM	79/48	-0.0003	-0.0007	0.0168	0.0175	[0.0126, 0.0232]
28	LR	80/48	0.0027	0.0056	0.0236	0.0180	[0.0118, 0.0251]
28	LightGBM	80/48	-0.0001	-0.0002	0.0121	0.0122	[0.0078, 0.0178]

TABLE S16. Baseline third-presentation full-Platt advantage on each budget's actual support. All pointwise intervals use 100,000 course resamples. C lists clusters at 10/20/50/100 percent; Day 7 sequences are 29/30/30/30, Days 14 and 28 are 32 at every budget. Figure 5 instead uses common support.

Day	Model	C by budget	10%	20%	50%	100%
7	LR	22/23/23/23	-0.0161 [-0.0309, -0.0003]	-0.0025 [-0.0146, 0.0116]	0.0037 [-0.0084, 0.0175]	0.0056 [-0.0063, 0.0194]
7	LightGBM	22/23/23/23	-0.0237 [-0.0387, -0.0075]	-0.0103 [-0.0204, 0.0021]	-0.0020 [-0.0118, 0.0098]	0.0012 [-0.0082, 0.0127]
14	LR	24/24/24/24	0.0046 [-0.0084, 0.0195]	0.0074 [-0.0064, 0.0229]	0.0120 [-0.0015, 0.0275]	0.0135 [0.0005, 0.0285]
14	LightGBM	24/24/24/24	-0.0052 [-0.0161, 0.0088]	-0.0001 [-0.0114, 0.0137]	0.0044 [-0.0061, 0.0173]	0.0059 [-0.0041, 0.0186]
28	LR	24/24/24/24	-0.0016 [-0.0129, 0.0103]	0.0008 [-0.0103, 0.0127]	0.0031 [-0.0081, 0.0148]	0.0037 [-0.0073, 0.0152]
28	LightGBM	24/24/24/24	-0.0071 [-0.0135, -0.0010]	-0.0039 [-0.0094, 0.0017]	-0.0016 [-0.0068, 0.0036]	-0.0010 [-0.0061, 0.0041]

evaluation remainder E. All three Platt calibrators use the same frozen scores and label count. We calculate temporal contrast $L_E(C_S) - L_E(C_C)$, random within-presentation contrast $L_E(C_H) - L_E(C_C)$, and their paired difference $L_E(C_S) - L_E(C_H)$. Neither target calibrator uses any E learner. This construction has no introduced temporal split between H and C; it does not assume the entire presentation is stationary over calendar time.[17]

Five random repetitions are attempted for each of 1,582 jobs. We require $m \geq 30$, evaluation N at least 30, and both classes with at least five observations in each calibration group. Of 7,910 attempts, 7,308 are supported; none is redrawn. Repetitions average within seed, then seeds within pair. The supported panel has 46/57/66 pairs and 196/259/312 source seeds per model. Pointwise intervals resample base courses. These split-sample estimands use a different evaluation subset from the primary OOF result and should not be substituted for it. Absolute prevalence, intercept and slope differences are also correlated with A descriptively in the machine-readable results; they do not identify causal mechanisms.

S18. CROSSED DEPENDENCE AND COURSE INFLUENCE

For each primary target learner within a pair, average the paired Brier-loss difference over fitted seeds. Draw independent exponential weights with mean and variance one for each observed user and each base course. A user's weight is shared across all pairs containing that user. Compute a user-weighted mean within each pair, then average pair means using their base-course weights. Repeat 2,000 times. This product-weight sensitivity follows the crossed-factor reweighting rationale cited in the main text, with within-pair normalization preserving the pair-macro estimand. Unit weights reproduce the primary mean to less than 10^{-12}.[18]

The resulting intervals condition on all saved predictors, calibrators and fold assignments. They reflect crossed dependence among scored losses but do not include dependence induced through refitting on shared users, uncertainty in course matching, or the effect of unobserved platform factors. They therefore complement, rather than replace, the one-base-course-per-user full refit. The leave-one-base-course-out range tests whether deleting a single observed cluster reverses the mean; it does not prove external validity.

S19. POST-CALIBRATION RELIABILITY DIAGNOSTICS

For raw, historical-Platt and target-OOF probabilities, the fixed-slope diagnostic intercept and bounded two-parameter diagnostic slope use all scored target outcomes. They do not alter those probabilities or select the models. Seed means are summarized by the median over pairs in Table S19.

[17] Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

[18] Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

TABLE S17. Equal-budget split-sample temporal and within-presentation contrasts. All calibrators have the same number of labels and share an independent evaluation subset. These are not the original OOF estimands.

Day	Model	Pairs	Temporal gap	Random gap	95% CI: random	Difference	95% CI: difference
7	LR	46	0.0406	-0.0010	[-0.0030, 0.0006]	0.0417	[0.0255, 0.0619]
7	LightGBM	46	0.0308	-0.0000	[-0.0012, 0.0010]	0.0308	[0.0162, 0.0514]
14	LR	57	0.0323	0.0001	[-0.0005, 0.0007]	0.0322	[0.0174, 0.0486]
14	LightGBM	57	0.0252	0.0000	[-0.0007, 0.0008]	0.0252	[0.0118, 0.0410]
28	LR	66	0.0169	0.0001	[-0.0002, 0.0005]	0.0168	[0.0091, 0.0253]
28	LightGBM	66	0.0147	-0.0001	[-0.0005, 0.0004]	0.0148	[0.0076, 0.0226]

TABLE S18. Conditional user–course reweighting and course influence. The interval holds all trained predictors and calibration mappings fixed. The final column is the range of leave-one-base-course-out point estimates, not a confidence interval.

Day	Model	Pairs	Courses	Users	A	Conditional 95% interval	Leave-one-course range
7	LR	48	37	16127	0.0412	[0.0266, 0.0618]	[0.0331, 0.0428]
7	LightGBM	48	37	16127	0.0307	[0.0186, 0.0514]	[0.0217, 0.0317]
14	LR	57	39	24921	0.0343	[0.0210, 0.0524]	[0.0287, 0.0354]
14	LightGBM	57	39	24921	0.0277	[0.0156, 0.0451]	[0.0216, 0.0287]
28	LR	66	40	35435	0.0188	[0.0117, 0.0287]	[0.0166, 0.0194]
28	LightGBM	66	40	35435	0.0166	[0.0106, 0.0252]	[0.0148, 0.0173]

TABLE S19. Calibration of raw, transferred, and target OOF probabilities: medians across seed-averaged pairs. Diagnostic intercept and slope fits use all scored target outcomes; they are descriptive and do not feed back into the evaluated predictions.

Day	Model	Raw int.	Raw slope	Hist. int.	Hist. slope	OOF int.	OOF slope	ECE raw / hist. / OOF
7	LR	0.755	0.202	0.643	0.598	0.001	0.633	0.167 / 0.133 / 0.059
7	LightGBM	0.459	0.692	0.487	0.776	0.000	0.753	0.110 / 0.125 / 0.057
14	LR	0.452	0.297	0.317	0.676	0.000	0.820	0.142 / 0.109 / 0.050
14	LightGBM	0.226	0.924	0.181	0.873	-0.000	0.930	0.090 / 0.093 / 0.044
28	LR	0.130	0.445	0.127	0.849	0.001	0.946	0.092 / 0.075 / 0.037
28	LightGBM	-0.003	0.930	0.099	0.917	0.000	0.975	0.068 / 0.070 / 0.036

ECE retains the original approximately equal-count ten-bin definition and its limitations. The target OOF intercept is near zero, but the early OOF slope remains below one; score improvement should not be equated with perfect reliability.[19]

Figure S1 uses ten fixed-width probability bins, preserves ties and omits empty bins. Among supported Day 7 LR seed-17 jobs, target N selects the smallest, middle-ranked and largest examples (pair-ID tie break); both models use these pairs and seed. Selection ignores calibration gain. Curves are descriptive without confidence bands; machine-readable bin counts are supplied.

S20. CONSERVATIVE UPDATING AND EQUAL-LABEL-BUDGET RECENCY

The conservative updating analysis is exploratory and was not preregistered. Strategy definitions, the count-based weight, budgets and comparison families were fixed after inspecting the full-Platt findings but before evaluating the conservative strategies. The baseline calculations reuse the saved calibration, T1 and T2 probabilities from 896 frozen source predictors. Reconstruction checks against the full-Platt rollout give maximum historical and full-update prediction discrepancies of zero and 7.22×10^{-16}, respectively. Reusing these predictions requires no additional base-model fits. Section S24 documents the raw-data execution used for the reported tables.[20]

Let $z_i = \mathrm{logit}(p_i)$ and $\theta_S = (a_S, b_S)$. The historical-slope intercept update holds a_S fixed and minimizes mean T1 logistic loss plus $10^{-5}(b - b_S)^2$, with $b \in [-30, 30]$. It therefore differs from the earlier raw-score intercept calibrator, which fixes slope to one. The full T1 fit retains the original Platt constraints and identity penalty. The shrinkage map has parameters $(1-w)\theta_S + w\theta_{T_1}$, with $w = n_{T_1}/(n_{T_1}+n_{S,cal})$. The count weight has no selected hyperparameter and is not an estimate of relative parameter precision. Probabilities retain 10^{-6} clipping. Neither function accesses T2 outcomes.

The same nested T1 permutations and support rules are retained for every strategy: at least ten labels and two observed classes. There are 10,752 strategy–budget evaluations, of which 10,722 are supported. A shrinkage combination is evaluated only where the full T1 update can be fitted; unsupported samples are not replaced. Within every job, strategies are scored on exactly the same T2 learners. Across budgets, the main figure restricts all curves to common support; Table

[19]Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

[20]Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

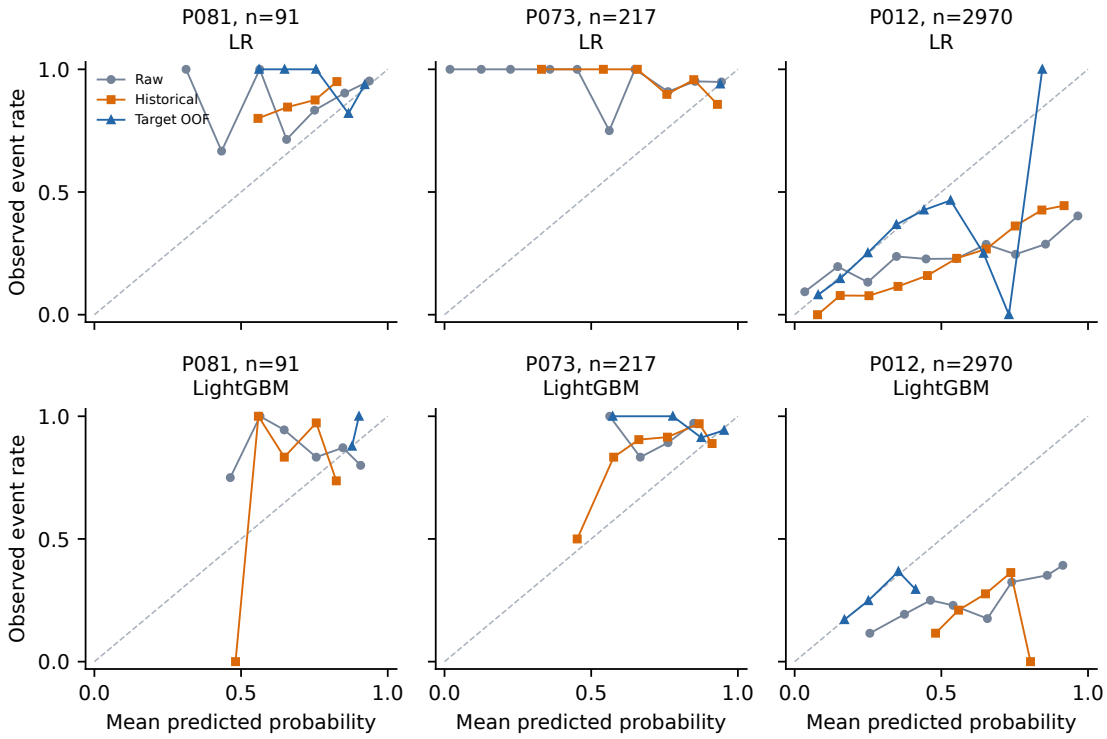


FIGURE S1. **Day 7 reliability examples selected by target sample size. Gray: raw; orange: historical Platt; blue: target OOF Platt. The diagonal denotes agreement between predicted probability and observed frequency. Bins are fixed by probability, with empty bins omitted.**

S20 uses each budget's actual supported jobs.

The matched recency comparison uses $m = \min(n_{S,cal}, n_{T_1})$ labels from each period. T1 follows the original permutation seeded by $20261011+\text{seed}+1009r+d$, where r is the numeric triple ID. The independent S permutation uses $2026101100 + \text{seed} + 1009r + d$. Both fits require $m \geq 10$ and both classes; no redraw is allowed. Full Platt is refitted separately to these subsets and evaluated on the identical T2 observations. There are 3,572 supported comparisons out of 3,584. Besides the ten previously unsupported Day 7 cases, one Day 28 seed per model fails the matched-S class condition at 10%. Matching controls label count, not other historical/recent differences, and is not a causal effect of recency.

At Day 7 and 10% labels, shrinkage reduces the magnitude of the negative mean advantages to 0.0001 (LR) and 0.0010 (LightGBM); its pointwise intervals still include zero. At full budget, shrinkage has the largest point estimate in all six model–landmark conditions. This ranking is descriptive, not evidence that shrinkage dominates. All six full-budget matched-budget intervals include zero. Complete absolute scores, fitted parameters, support counts and all per-budget intervals are supplied as aggregate tables, without learner records.

S21. MULTIPLICITY SENSITIVITY AND INDEPENDENT SUPPORT

Random seeds, budgets and strategies are repeated calculations, not additional independent courses. The full-budget third-presentation panel has 23/24/24 base-course clusters and 30/32/32 triples at Days 7/14/28. The 10% Day 7 budget supports 22 clusters and 29 triples. Multiple triples within a base course are resampled together; a supported source split is averaged within its triple before equal-weight triple averaging.[21]

We use 100,000 cluster resamples with random seed 20261012. A draw samples C base courses with replacement, includes all their supported triples and recomputes the triple-macro mean. For a comparison family of size K, percentile endpoints are evaluated at $\alpha/(2K)$ and $1 - \alpha/(2K)$, with $\alpha = 0.05$. This Bonferroni expansion uses the union-bound rationale; the individual bootstrap intervals have only approximate coverage with few clusters, so exact simultaneous coverage is not claimed. These calculations condition on fitted predictors, calibrators and selected support; shared-user fitting dependence is not removed.

The focused family has K=6 full-budget full-Platt contrasts. Their expanded intervals all include zero. In particular, Day 14 LR has a pointwise interval of [0.0005, 0.0285] and a six-comparison interval of [-0.0033, 0.0345], both computed using 100,000 course-cluster draws.

[21] Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

TABLE S20. Baseline update advantage: mean [pointwise 95% interval] from 100,000 cluster resamples. All three strategies share actual per-budget support (Table S16). All 72 baseline strategy contrasts have zero-crossing intervals under the baseline 96-comparison family; this statement does not apply to the restricted populations.

Day	Model	Strategy	10%	20%	50%	100%
7	LR	full	-0.0161 [-0.0309, -0.0003]	-0.0025 [-0.0146, 0.0116]	0.0037 [-0.0084, 0.0175]	0.0056 [-0.0063, 0.0194]
7	LR	intercept	-0.0100 [-0.0197, 0.0015]	-0.0042 [-0.0145, 0.0076]	-0.0009 [-0.0113, 0.0111]	0.0009 [-0.0089, 0.0126]
7	LR	shrink	-0.0001 [-0.0081, 0.0106]	0.0036 [-0.0042, 0.0139]	0.0059 [-0.0022, 0.0165]	0.0076 [-0.0013, 0.0187]
7	LightGBM	full	-0.0237 [-0.0387, -0.0075]	-0.0103 [-0.0204, 0.0021]	-0.0020 [-0.0118, 0.0098]	0.0012 [-0.0082, 0.0127]
7	LightGBM	intercept	-0.0116 [-0.0220, 0.0014]	-0.0060 [-0.0158, 0.0061]	-0.0025 [-0.0123, 0.0092]	-0.0003 [-0.0092, 0.0110]
7	LightGBM	shrink	-0.0010 [-0.0086, 0.0094]	0.0027 [-0.0029, 0.0116]	0.0035 [-0.0028, 0.0129]	0.0045 [-0.0025, 0.0143]
14	LR	full	0.0046 [-0.0084, 0.0195]	0.0074 [-0.0064, 0.0229]	0.0120 [-0.0015, 0.0275]	0.0135 [0.0005, 0.0285]
14	LR	intercept	0.0073 [-0.0044, 0.0205]	0.0092 [-0.0029, 0.0230]	0.0113 [-0.0001, 0.0247]	0.0121 [0.0009, 0.0253]
14	LR	shrink	0.0093 [0.0025, 0.0194]	0.0119 [0.0038, 0.0232]	0.0152 [0.0054, 0.0276]	0.0163 [0.0057, 0.0292]
14	LightGBM	full	-0.0052 [-0.0161, 0.0088]	-0.0001 [-0.0114, 0.0137]	0.0044 [-0.0061, 0.0173]	0.0059 [-0.0041, 0.0186]
14	LightGBM	intercept	-0.0014 [-0.0115, 0.0109]	0.0014 [-0.0089, 0.0136]	0.0037 [-0.0059, 0.0155]	0.0043 [-0.0049, 0.0159]
14	LightGBM	shrink	0.0056 [-0.0008, 0.0156]	0.0070 [0.0000, 0.0179]	0.0084 [0.0008, 0.0196]	0.0088 [0.0007, 0.0201]
28	LR	full	-0.0016 [-0.0129, 0.0103]	0.0008 [-0.0103, 0.0127]	0.0031 [-0.0081, 0.0148]	0.0037 [-0.0073, 0.0152]
28	LR	intercept	0.0021 [-0.0073, 0.0118]	0.0034 [-0.0055, 0.0126]	0.0040 [-0.0053, 0.0134]	0.0043 [-0.0047, 0.0135]
28	LR	shrink	0.0049 [0.0004, 0.0111]	0.0063 [0.0007, 0.0137]	0.0073 [-0.0001, 0.0163]	0.0071 [-0.0012, 0.0167]
28	LightGBM	full	-0.0071 [-0.0135, -0.0010]	-0.0039 [-0.0094, 0.0017]	-0.0016 [-0.0068, 0.0036]	-0.0010 [-0.0061, 0.0041]
28	LightGBM	intercept	-0.0042 [-0.0102, 0.0018]	-0.0026 [-0.0079, 0.0028]	-0.0014 [-0.0068, 0.0039]	-0.0014 [-0.0067, 0.0039]
28	LightGBM	shrink	0.0014 [-0.0007, 0.0044]	0.0017 [-0.0008, 0.0050]	0.0016 [-0.0016, 0.0055]	0.0012 [-0.0025, 0.0053]

TABLE S21. Baseline equal-label-budget S/T1 recency comparison: matched-S Brier minus matched-T1 Brier, with 100,000-draw pointwise intervals. Support follows Table S20 except one extra source split per model fails at Day 28, 10%. All full-budget pointwise and all 24 baseline 96-family intervals cross zero.

Day	Model	10%	20%	50%	100%
7	LR	-0.0002 [-0.0126, 0.0136]	0.0018 [-0.0112, 0.0164]	0.0022 [-0.0110, 0.0171]	0.0017 [-0.0116, 0.0166]
7	LightGBM	-0.0098 [-0.0236, 0.0051]	-0.0080 [-0.0186, 0.0044]	-0.0036 [-0.0142, 0.0086]	-0.0034 [-0.0141, 0.0090]
14	LR	0.0133 [-0.0010, 0.0285]	0.0115 [-0.0027, 0.0262]	0.0103 [-0.0029, 0.0251]	0.0108 [-0.0021, 0.0254]
14	LightGBM	0.0006 [-0.0101, 0.0135]	0.0016 [-0.0103, 0.0151]	0.0028 [-0.0082, 0.0157]	0.0032 [-0.0075, 0.0160]
28	LR	0.0031 [-0.0082, 0.0149]	0.0010 [-0.0099, 0.0125]	0.0032 [-0.0075, 0.0146]	0.0032 [-0.0075, 0.0146]
28	LightGBM	-0.0029 [-0.0096, 0.0032]	-0.0036 [-0.0094, 0.0020]	-0.0022 [-0.0071, 0.0027]	-0.0019 [-0.0067, 0.0028]

The extended sensitivity defines K=96 before evaluating the conservative strategies: 72 historical-map contrasts (three updating strategies, two models, three landmarks and four budgets) plus 24 matched S/T1 contrasts. All 96 baseline-population expanded intervals include zero; the recent-active and joint-assigned populations have the four scoped signals in Table S27. They are available in the machine-readable update_intervals.csv; Table S22 illustrates the same expansion for the original six full-budget contrasts. This sensitivity is applied separately within each population. It does not cover all archival-label and time-defined analyses jointly; it cannot provide a global family-wise interpretation. A null-crossing interval neither proves updating ineffective nor establishes equivalence. Strategy selection would require a separately specified policy and new independent evaluation.

S22. RECENT ACTIVITY AND JOINT ASSIGNMENT: FULL-PIPELINE POPULATIONS

The population-sensitivity protocol was fixed before fitting these analyses, after inspecting the baseline findings; it was not preregistered. Both restricted populations use the structural 81-pair pool and the time-defined outcome in S15. In the recent-active analysis, last observed activity must exceed day $d - 7$, in addition to having an event by d. Original full-enrollment, outcome-stratified source role allocation precedes this restriction; vocabulary, preprocessing, base models and all calibration fits are then recomputed on the retained source roles. The same risk-set rule applies to T1 and T2. Recent-active Day 7 coincides with baseline on supported jobs and is not another independent replication.[22]

Joint assignment uses the earliest observed event to assign each username to one base course within the 129 presentations and 48 base courses involved in the 81 structural pairs, with lexical tie breaking. It is not a platform-wide allocation. Before allocation, 36,988 of 63,600 users occur across multiple eligible base courses; afterwards none do. Eligible enrollments decrease from 131,195 to 65,400. All source roles, features, vocabularies, predictors and calibrators are rebuilt after allocation. The complete original enrollment rosters, including learners excluded from fitting, enforce source-user exclusion from T1 and S/T1-user exclusion from T2. This restriction does not remove every possible course dependence. The recent-active and joint-assigned analyses are separate sensitivities, not their intersection.

[22]Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

TABLE S22. Baseline full-budget full-Platt U under pointwise, six-comparison and 96-comparison percentile expansions, all from 100,000 course resamples. Families concern this baseline population only. Table S26 extends the six-comparison display to the other populations.

Day	Model	C	U	Pointwise	Six-comparison	96-comparison
7	LR	23	0.0056	[-0.0063, 0.0194]	[-0.0103, 0.0247]	[-0.0148, 0.0319]
7	LightGBM	23	0.0012	[-0.0082, 0.0127]	[-0.0111, 0.0175]	[-0.0146, 0.0243]
14	LR	24	0.0135	[0.0005, 0.0285]	[-0.0033, 0.0345]	[-0.0076, 0.0422]
14	LightGBM	24	0.0059	[-0.0041, 0.0186]	[-0.0070, 0.0240]	[-0.0100, 0.0313]
28	LR	24	0.0037	[-0.0073, 0.0152]	[-0.0112, 0.0193]	[-0.0160, 0.0245]
28	LightGBM	24	-0.0010	[-0.0061, 0.0041]	[-0.0078, 0.0059]	[-0.0100, 0.0083]

The source training/validation minority thresholds are five/two. Calibration requires at least 30 observations and ten in each class; later evaluation requires 50 observations and ten per class. Both observed spans must cover $d + 7$. Update samples require ten labels and both classes, without redrawing failed samples. Table S23 distinguishes supported units, clusters and repeated fits. Common-job comparisons match pair or sequence, day, seed and model; they do not equate learner membership or isolate a causal refitting effect.

S23. AGGREGATE EVIDENCE AND CONDITIONAL UNCERTAINTY

The accompanying evidence registry records source CSV hashes, exact row keys, full precision means, intervals, support and comparison families. The main-text population and updating tables are generated from these frozen sources. Table S24 gives the additional pair decompositions; Tables S25–S26 give all full-budget absolute performance and paired full-update uncertainty. The 12-comparison family spans the recent-active and joint-assigned populations' A contrasts. Each population separately has a six-comparison full-Platt family and a 96-comparison extended updating family. No global correction across all historical and exploratory analyses is claimed.[23]

All time-defined population and updating summaries use 100,000 cluster resamples, seed 20261012. Original archival-label analyses retain their 2,000-draw protocol. The bootstrap conditions on saved fits and support, omitting preprocessing, predictor-fitting, calibration-fitting and selection uncertainty. This remains true even though the population estimates were generated with full-pipeline refits. The joint-assigned Day 7 rollout has eight clusters, making tail inference especially fragile. Null-crossing intervals establish neither ineffectiveness nor equivalence.

Figure 5 recomputes both means and pointwise intervals from the intersection of model jobs supported across all four budgets and three strategies. Its Day 7 support is 29 sequences/22 clusters, versus full-budget 30/23. No 96-comparison marker from an actual-budget support set is attached to that common panel. Figure S4 instead uses the recent-active actual support separately at each budget; its stars refer only to that population's 96-comparison family. Complete per-budget N and C accompany the machine-readable interval rows.

Table S27 reports all four population-specific tail signals with primary 100,000-draw endpoints. One-million-draw resampling with seed 20261014 checks only their Monte Carlo stability, without changing the final inferential display or expanding the tested strategy set. Three favorable recent-active LR shrinkage conditions have lower bounds close to zero and 20–24 clusters. The negative joint Day 14 LightGBM 10% full-update condition has 14 clusters. The four tail directions survive this resampling check, which is not independent data validation or a globally adjusted strategy-selection guarantee.

Table S28 and Fig. S3 report future-update course heterogeneity. Leave-one-course-out means remain conditional on the existing fits; no course is dropped to improve a result. Positive A across these restricted analyses does not guarantee positive later U. This comparison is not a formal test that A cannot statistically predict U.

S24. EXECUTED RAW-DATA CHECKS AND REPRODUCTION BOUNDARIES

The fresh run reads the original 379,666,118-byte XuetangX archive (SHA-256 beginning 457683ea9e540b58; complete value in the protocol). It reconstructs 42,110,402 events, 225,642 enrollments and 247 presentations. All preparation counts and structural support keys match the prior execution. Preparation feature differences are zero except standard deviations within 1.82×10^{-12}. The baseline pair and rollout reruns reproduce keys, support and model settings; maximum absolute per-job differences in A and U are 3.33×10^{-7} and 4.38×10^{-7}, respectively. Across the 96 baseline summary comparisons, maximum numeric difference is 1.46×10^{-8}. Calibration parameters can differ by up to 4.45×10^{-5}; bitwise model identity is not asserted.[24]

The run comprises baseline 2,186 pair plus 896 rollout fits; recent-active 1,932 plus 828; and joint-assigned 1,256 plus 512, totaling 7,610 base-model fits. All 7,610 saved prediction artifacts were checked for role separation, source/later learner exclusions, appropriate risk sets and time boundaries; no audited violation was found. These are author-executed audits of implemented constraints, not independent assurance of complete logs, authentic labels or all possible leakage pathways. Equal numerical results do not constitute independent-

[23]Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

[24]Drafting assistance: OpenAI Codex [1]; see the main Acknowledgment.

TABLE S23. full-pipeline support by population and landmark. Pairs and sequences use the explicitly timed inactivity outcome. Update evaluations count three strategies and four budgets, not independent cohorts. Day 7 recent-active jobs duplicate baseline support.

Population	Day	Pairs/clusters	Pair fits	Sequences/clusters	Rollout fits	Supported/attempted updates
Baseline	7	73/47	678	30/23	282	3354/3384
Baseline	14	79/48	736	32/24	304	3648/3648
Baseline	28	80/48	772	32/24	310	3720/3720
Recent-active	7	73/47	678	30/23	282	3354/3384
Recent-active	14	75/47	668	31/24	276	3282/3312
Recent-active	28	67/43	586	29/23	270	3078/3240
Joint-assigned	7	34/28	298	10/8	88	990/1056
Joint-assigned	14	53/41	466	23/19	210	2298/2520
Joint-assigned	28	56/42	492	26/21	214	2502/2568

TABLE S24. Full gain decomposition for the two restricted populations; 100,000 course-cluster resamples. The 12-comparison adjustment applies only to A across these rows; support conditions and population overlap preclude interpreting the rows as independent replications.

Population	Day	Model	Pairs/C	G_S	G_T	G_T^*	A	Adjusted interval
Recent-active	7	LR	73/47	0.0108	0.0185	0.0447	0.0261	[0.0159, 0.0400]
Recent-active	7	LightGBM	73/47	-0.0005	0.0016	0.0190	0.0174	[0.0096, 0.0286]
Recent-active	14	LR	75/47	0.0120	0.0141	0.0422	0.0280	[0.0181, 0.0405]
Recent-active	14	LightGBM	75/47	-0.0006	-0.0019	0.0208	0.0228	[0.0138, 0.0341]
Recent-active	28	LR	67/43	0.0085	0.0144	0.0360	0.0216	[0.0112, 0.0346]
Recent-active	28	LightGBM	67/43	0.0006	-0.0010	0.0152	0.0163	[0.0091, 0.0265]
Joint-assigned	7	LR	34/28	0.0116	0.0218	0.0456	0.0238	[0.0139, 0.0368]
Joint-assigned	7	LightGBM	34/28	0.0008	-0.0064	0.0110	0.0174	[0.0086, 0.0273]
Joint-assigned	14	LR	53/41	0.0108	0.0240	0.0554	0.0314	[0.0169, 0.0521]
Joint-assigned	14	LightGBM	53/41	-0.0009	-0.0012	0.0159	0.0171	[0.0100, 0.0260]
Joint-assigned	28	LR	56/42	0.0014	0.0104	0.0279	0.0175	[0.0097, 0.0272]
Joint-assigned	28	LightGBM	56/42	-0.0009	0.0031	0.0156	0.0125	[0.0069, 0.0196]

TABLE S25. All 18 full-budget absolute T2 performance conditions. Equal sequence weights follow within-sequence seed averaging. The constant uses T1 labels; T2 prevalence is only descriptive. Pointwise 100,000-draw intervals for every absolute score are supplied in the machine-readable registry, with paired U intervals in Table S26.

Population	Day	Model	N/C	T2 rate	Raw	Historical	T1 constant	Full	Intercept	Shrink
Baseline	7	LR	30/23	0.6416	0.2435	0.2238	0.2389	0.2182	0.2229	0.2162
Baseline	7	LightGBM	30/23	0.6416	0.2129	0.2108	0.2389	0.2095	0.2111	0.2063
Baseline	14	LR	32/24	0.7338	0.2003	0.1878	0.1998	0.1744	0.1757	0.1715
Baseline	14	LightGBM	32/24	0.7338	0.1712	0.1726	0.1998	0.1666	0.1682	0.1638
Baseline	28	LR	32/24	0.7797	0.1569	0.1485	0.1747	0.1447	0.1442	0.1414
Baseline	28	LightGBM	32/24	0.7797	0.1334	0.1345	0.1747	0.1355	0.1359	0.1333
Recent-active	7	LR	30/23	0.6416	0.2435	0.2238	0.2389	0.2182	0.2229	0.2162
Recent-active	7	LightGBM	30/23	0.6416	0.2129	0.2108	0.2389	0.2095	0.2111	0.2063
Recent-active	14	LR	31/24	0.6667	0.2369	0.2192	0.2304	0.2054	0.2068	0.2014
Recent-active	14	LightGBM	31/24	0.6667	0.1996	0.2010	0.2304	0.1945	0.1953	0.1905
Recent-active	28	LR	29/23	0.6140	0.2364	0.2203	0.2437	0.2185	0.2161	0.2128
Recent-active	28	LightGBM	29/23	0.6140	0.2054	0.2061	0.2437	0.2069	0.2044	0.2031
Joint-assigned	7	LR	10/8	0.6266	0.2396	0.2196	0.2399	0.2116	0.2187	0.2081
Joint-assigned	7	LightGBM	10/8	0.6266	0.2051	0.2048	0.2399	0.1991	0.2048	0.1967
Joint-assigned	14	LR	23/19	0.7307	0.2132	0.1933	0.2030	0.1820	0.1896	0.1813
Joint-assigned	14	LightGBM	23/19	0.7307	0.1727	0.1756	0.2030	0.1727	0.1767	0.1704
Joint-assigned	28	LR	26/21	0.7446	0.1739	0.1631	0.1887	0.1620	0.1654	0.1589
Joint-assigned	28	LightGBM	26/21	0.7446	0.1510	0.1530	0.1887	0.1519	0.1549	0.1500

team replication. Original-label XuetangX and all OULAD workflows remain outside the scope of this fresh full execution.

The reproducibility workflow distinguishes (i) checked regeneration of frozen aggregates, (ii) the documented raw-data execution above, and (iii) broader workflows provided for other analyses without claiming a complete raw-data rerun. Protocols, source hashes and executed validation records document the analysis. The source and aggregate identifiers in Section S13 identify the analysis materials; raw-data exe-

TABLE S26. Full-budget full-Platt future update advantage U. Intervals use 100,000 base-course resamples. Six-comparison adjustments are applied separately within each population, not jointly to all 18 rows or all paper analyses. Every adjusted interval crosses zero.

Population	Day	Model	C	*U*	Pointwise 95% interval	Population-wise six-comparison interval
Baseline	7	LR	23	0.0056	[-0.0063, 0.0194]	[-0.0103, 0.0247]
Baseline	7	LightGBM	23	0.0012	[-0.0082, 0.0127]	[-0.0111, 0.0175]
Baseline	14	LR	24	0.0135	[0.0005, 0.0285]	[-0.0033, 0.0345]
Baseline	14	LightGBM	24	0.0059	[-0.0041, 0.0186]	[-0.0070, 0.0240]
Baseline	28	LR	24	0.0037	[-0.0073, 0.0152]	[-0.0112, 0.0193]
Baseline	28	LightGBM	24	-0.0010	[-0.0061, 0.0041]	[-0.0078, 0.0059]
Recent-active	7	LR	23	0.0056	[-0.0063, 0.0194]	[-0.0103, 0.0247]
Recent-active	7	LightGBM	23	0.0012	[-0.0082, 0.0127]	[-0.0111, 0.0175]
Recent-active	14	LR	24	0.0138	[-0.0006, 0.0305]	[-0.0050, 0.0370]
Recent-active	14	LightGBM	24	0.0065	[-0.0044, 0.0194]	[-0.0078, 0.0246]
Recent-active	28	LR	23	0.0018	[-0.0080, 0.0116]	[-0.0112, 0.0148]
Recent-active	28	LightGBM	23	-0.0008	[-0.0070, 0.0053]	[-0.0093, 0.0072]
Joint-assigned	7	LR	8	0.0080	[-0.0068, 0.0256]	[-0.0111, 0.0316]
Joint-assigned	7	LightGBM	8	0.0056	[-0.0040, 0.0186]	[-0.0067, 0.0235]
Joint-assigned	14	LR	19	0.0113	[-0.0033, 0.0251]	[-0.0087, 0.0294]
Joint-assigned	14	LightGBM	19	0.0029	[-0.0048, 0.0105]	[-0.0075, 0.0126]
Joint-assigned	28	LR	21	0.0011	[-0.0082, 0.0118]	[-0.0114, 0.0161]
Joint-assigned	28	LightGBM	21	0.0011	[-0.0032, 0.0062]	[-0.0046, 0.0081]

TABLE S27. Four exploratory tail signals. The 96-comparison family is defined separately within each population; these are not a global correction across all analyses. Primary endpoints use 100,000 draws. One million draws only check Monte Carlo stability of the four selected tails. Small cluster counts and near-zero lower bounds limit positive-strategy claims.

Population	Day	Model	Budget	Strategy	C	*U*	100k adjusted interval	1M stability interval
Recent-active	14	LR	20%	shrink	24	0.0136	[0.0009, 0.0406]	[0.0009, 0.0398]
Recent-active	28	LR	10%	shrink	20	0.0064	[0.0004, 0.0209]	[0.0004, 0.0203]
Recent-active	28	LR	20%	shrink	23	0.0071	[0.0003, 0.0182]	[0.0004, 0.0187]
Joint-assigned	14	LightGBM	10%	full	14	-0.0353	[-0.0995, -0.0023]	[-0.1005, -0.0024]

TABLE S28. Full-budget full-Platt course heterogeneity and leave-one-course-out (LOCO) means. Within-course means are used only for the distribution; overall U remains sequence weighted. LOCO ranges are influence summaries, not confidence intervals or post-hoc exclusion rules.

Population	Day	Model	C	Overall *U*	Positive courses	Course *U* range	LOCO mean range
Baseline	7	LR	23	0.0056	56.5%	[-0.0507, 0.1101]	[0.0020, 0.0096]
Baseline	7	LightGBM	23	0.0012	52.2%	[-0.0346, 0.1072]	[-0.0024, 0.0038]
Baseline	14	LR	24	0.0135	62.5%	[-0.0355, 0.1486]	[0.0091, 0.0168]
Baseline	14	LightGBM	24	0.0059	66.7%	[-0.0309, 0.1370]	[0.0017, 0.0084]
Baseline	28	LR	24	0.0037	66.7%	[-0.0538, 0.0864]	[0.0011, 0.0070]
Baseline	28	LightGBM	24	-0.0010	66.7%	[-0.0262, 0.0396]	[-0.0023, 0.0007]
Recent-active	7	LR	23	0.0056	56.5%	[-0.0507, 0.1101]	[0.0020, 0.0096]
Recent-active	7	LightGBM	23	0.0012	52.2%	[-0.0346, 0.1072]	[-0.0024, 0.0038]
Recent-active	14	LR	24	0.0138	66.7%	[-0.0394, 0.1604]	[0.0090, 0.0175]
Recent-active	14	LightGBM	24	0.0065	62.5%	[-0.0420, 0.1207]	[0.0027, 0.0099]
Recent-active	28	LR	23	0.0018	65.2%	[-0.0357, 0.0517]	[0.0000, 0.0045]
Recent-active	28	LightGBM	23	-0.0008	47.8%	[-0.0359, 0.0268]	[-0.0018, 0.0011]
Joint-assigned	7	LR	8	0.0080	75.0%	[-0.0218, 0.0528]	[0.0030, 0.0155]
Joint-assigned	7	LightGBM	8	0.0056	75.0%	[-0.0123, 0.0457]	[0.0012, 0.0101]
Joint-assigned	14	LR	19	0.0113	84.2%	[-0.0533, 0.0760]	[0.0083, 0.0174]
Joint-assigned	14	LightGBM	19	0.0029	73.7%	[-0.0240, 0.0309]	[0.0016, 0.0054]
Joint-assigned	28	LR	21	0.0011	66.7%	[-0.0384, 0.0846]	[-0.0022, 0.0042]
Joint-assigned	28	LightGBM	21	0.0011	47.6%	[-0.0147, 0.0359]	[-0.0003, 0.0023]

cution and aggregate-only verification remain separate reproducibility levels.

REFERENCES

[1] OpenAI, "Codex," AI-assisted coding and writing system, accessed: Oct. 2, 2026. [Online]. Available: https://openai.com/codex/

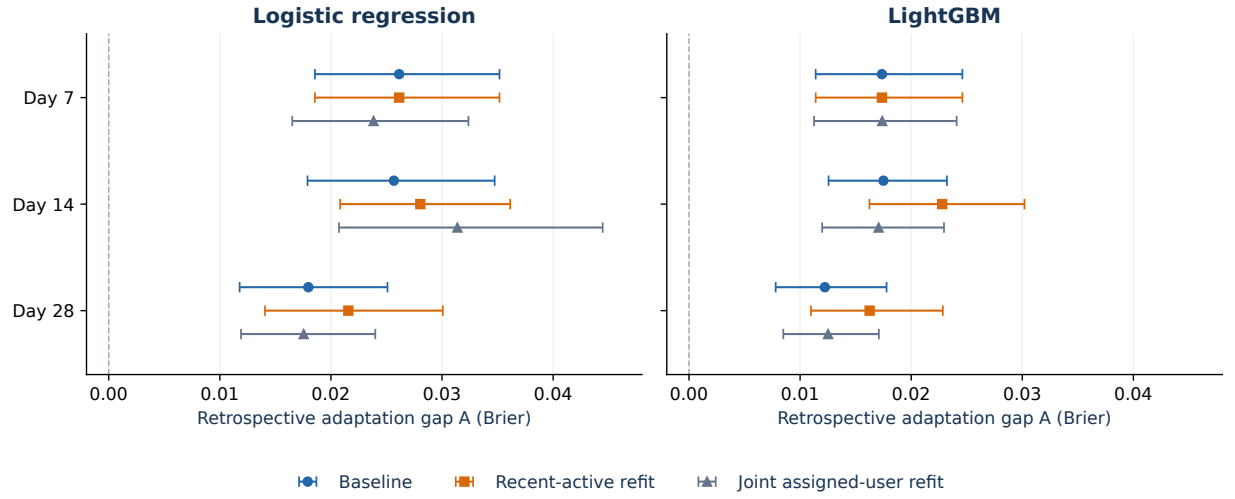


FIGURE S2. **Time-defined adaptation gaps across the three overlapping populations. Bars are 100,000-draw pointwise course-cluster intervals, distinct from the 12-comparison intervals in Table S24. Recent-active Day 7 repeats baseline.**

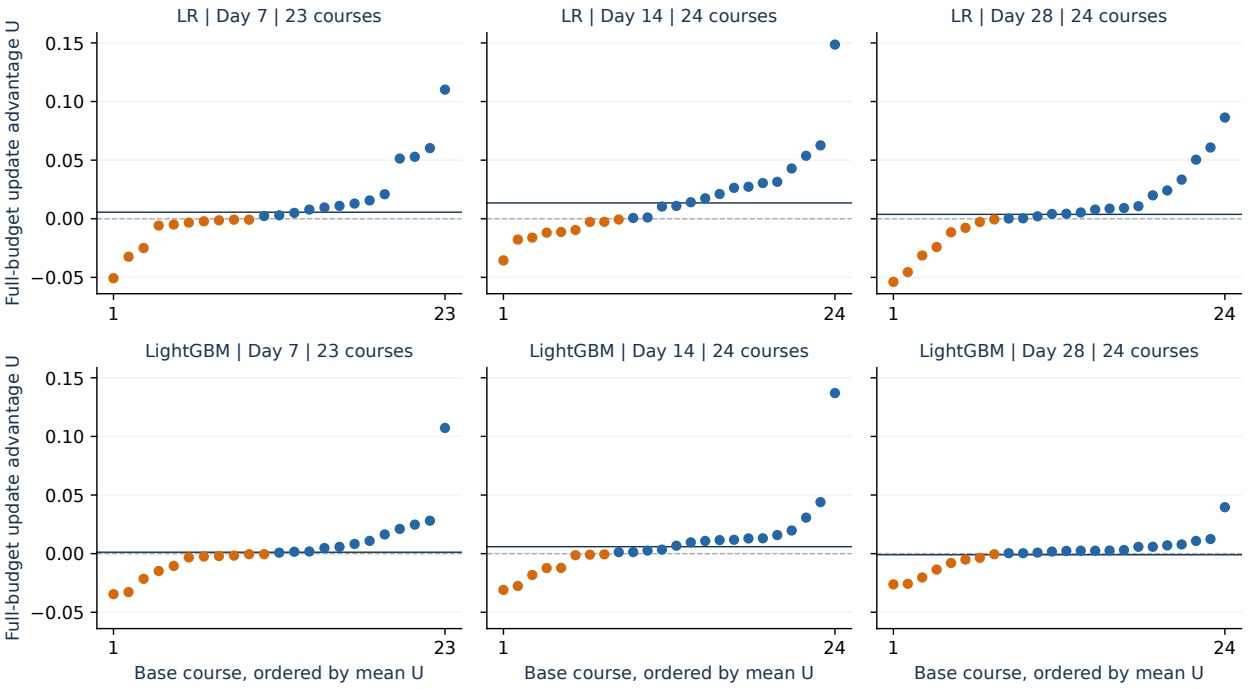


FIGURE S3. **Baseline full-budget full-Platt course means, sorted within each condition. The horizontal solid line is the sequence-weighted overall mean, not the unweighted mean of course points. Courses are displayed for heterogeneity assessment, not selected for exclusion.**

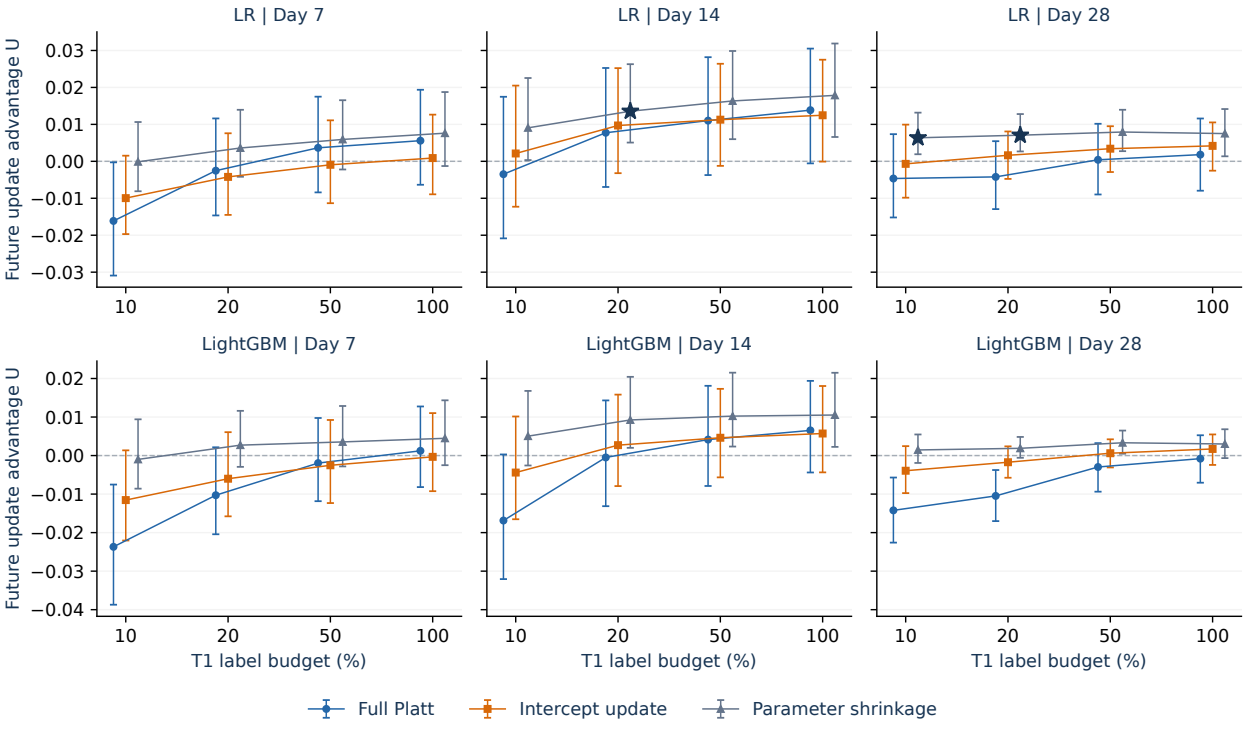


FIGURE S4. **Recent-active updating on each budget's actual supported jobs. Bars are pointwise 100,000-draw intervals; stars mark nonzero intervals under this population's 96-comparison expansion. Support may differ across budgets, so the plotted changes combine budget and composition differences; these markers do not apply to the common-support main Fig. 5. See interval rows for N/C at each budget.**